DIAFENTIURON GENBRA 500 SC

Material Safety Data Sheet

1. IDENTIFICATION

- Product name: Diafenthiuron Genbra 500 SC
- Main recommended uses: Acaricide-insecticide of contact and ingestion of the chemical group Phenylthiourea.
- Supplier: **GENBRA DISTRIBUIÇÃO DE PRODUTOS AGRÍCOLAS LTDA.**
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  Telephone/Fax: (41) 3019-0701
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2. HAZARD IDENTIFICATION

- **Most important hazards:** The product may be toxic to the environment if not used as recommended.
- **Product Effects**:
  Adverse effects to human health: the product is harmful if ingested or inhaled. May be harmful in contact with skin.
  Environmental effects: Very toxic to aquatic organisms.
  Physical and chemical hazards: no known hazards.

- **Primary Symptoms**: Ingestion of large amounts of the product may cause symptoms such as nausea, vomiting, colic and diarrhea. Direct contact with eyes or skin may cause redness and discomfort at the point of contact.

- **Hazard classification of the product:**
  System of classification of danger according to the Globally Harmonized Classification and Labeling of Chemicals System.
  ABNT-NBR 14725 – Part 2:2009.

  Acute toxicity - Oral: Category 4.
  Acute toxicity – Skin: Category 5.

Acute toxicity - Inhalation: Category 4.
Corrosion / Irritation to skin: Not classified.
Serious eye damage / eye irritation: Not classified.
Respiratory sensitizers: Classification not possible.
Skin Sensitization: Classification not possible.
Germ cell mutagenicity: Classification not possible.
Carcinogenicity: Classification not possible.
Reproductive toxicity: Classification not possible.
Systemic toxicity to target organ (single exposure): Classification not possible.
Target Organ Systemic Toxicity (repeated exposure): Classification not possible.
Aspiration Hazard: Classification not possible.
Hazardous to the aquatic environment - Acute: Category 1.
Danger to aquatic environment - Chronic: Classification not possible.
Flammable liquids: Classification not possible.
Corrosivity: Not classified.

- **Proper elements for labeling**:

Pictogram		
Warning word	Attention	

Hazard statements:
H302 - Harmful if swallowed.
H313 - May be harmful in contact with skin.
H332 - Harmful if inhaled.
H400 - Very toxic to aquatic organisms.

Precautionary statements:
P261 - Avoid inhalation of dust / fume / gas / mist / vapors / spray.
P270 - Do not eat, drink or smoke while using this product.
P271 - Use only in outdoors or in a well ventilated area.
P273 - Avoid release the product to the environment.
P391 - Collect spilled material.

3. COMPOSITION AND INFORMATION ABOUT INGREDIENTS

Chemical Nature: This product is a mixture.
Ingredients or impurities that contribute to the hazard:

Chemical Name	CAS Nº	Concentration	Molecular Formula	Synonyms	Hazard Classification
1-terc-butyl-3-(2,6-Diisopropyl-4-Fenoxifenyl) Thiourea	80060-09-9	500 g/L	$C_{23}H_{32}N_2OS$	Diafenthiuron	**Acute toxicity-Oral**: Category 5 **Acute toxicity-Skin:** Category 5 **Environmental hazard:** Category 3 **Danger to aquatic environment-acute:** Category 1
1,2-Etanodiol	107-21-1	50 g/L	$C_2H_6O_2$	Ethylenglycol	**Acute toxicity-Oral**: Category 5 **Acute toxicity-Inhalatory:** Category 5 **Corrosivity/Skin irritation:** Category 3 **Severe eyes lesions/eye irritation:** Category 2B

System of classification of danger according to the Globally Harmonized Classification and Labeling of Chemicals System.
ABNTNBR14725 - Part 2: 2009.

4. FIRST AID ACTIONS

- **First aid measures**: Remove to fresh air. Remove the contaminated clothing. Wash off affected parts of the body with plenty of water and soap. If the victim is unconscious and does not breathe, perform oxygenation or artificial respiration techniques. Refer to the nearest medical service.
- **Inhalation**: Remove to fresh air. If breathing is difficult, consult a physician immediately. If the victim is not breathing, perform artificial respiration techniques.
- **Skin contact**: Immediately wash off affected areas with plenty of water and soap. Remove contaminated clothing. If symptoms or effects occur, consult a physician. Wash off contaminated clothing before reuse.
- **Eye contact**: wash off immediately with plenty of water for at least 15 minutes or the maximum amount of time possible. Keep eyelids open in order to ensure proper eye rinse. Consult a physician if irritation develops.
- **Ingestion**: do not induce vomiting, however it is likely to occur spontaneously and should not be avoided. Lay the patient on his side to prevent aspiration. Get medical attention immediately. ATTENTION: never give something for orally to an unconscious person.
- **What actions should be avoided**: do not apply mouth-to-mouth to a person that has ingested the product.
- **Protection for first-aiders**: avoid oral, cutaneous, ocular and inhalation of the product during the administration of fist care techniques.
- **Notes to physician**: There is no specific antidote. In case of recent ingestion of large amounts, it is recommended to perform gastric lavage procedures and the administration of activated carbon. Symptomatic treatment should include correction of hydro electrolytic, metabolic and if necessary, monitoring of hepatic and renal functions. In case of contact with the eyes, wash off with saline solution.

5. FIRE FIGHTING MEASURES

Suitable extinguishing media: foam, dry chemical, CO_2 and water fog.

- **Extinguishing media not recommended**: avoid the use of water spray.
- **Specific hazards and special methods of firefighting**: evacuate the area and fight the fire at a safe distance. Use dikes to contain the water. Stand with your back to the wind. Use water spray to cool equipment near the fire. Keep the containers cooled with water spray.
- **Protection of personnel involved in firefighting**: breathing apparatus and appropriate fire-fighting clothing.
- **Specific hazards of combustion of the product**: combustion of the product may produce toxic or irritant fumes.

6. ACCIDENTAL RELEASE MEASURES

- **Personal precautions:** Wear waterproof overalls, goggles, rubber and gloves of nitrile rubber or PVC. Respiratory protection should be used depending on the concentrations present in the environment or the extent of spill / leakage. In this case, use semi-facial masks or facials with replaceable filter or even air-intake respirators (e.g. masks).
  Removal of ignition sources: discontinue electric power and turn off sources or generators. Remove any material that may cause fire to start (e.g. diesel oil).
  Prevention of inhalation and contact with skin, mucous membranes and eyes: use clothing and accessories described above under Personal Precautions.
- **Environmental precautions**: avoid contamination of sewers avoiding the entrance of rainwater. Prevent product residues from spills reaches collections of water.
- **Methods for cleaning up**: Eliminate all sources of fire or heat. Avoid contact with skin and clothing. **Floor pavement**: absorb the product with sand or sawdust, collect the material with the aid of a shovel, place in sealed container, properly identified. **Ground**: remove contaminated soil layers until reaching uncontaminated soil; collect this material and place in a sealed and properly labeled container. Contact the registrant.

Bodies of water: immediately stop the human or animal consumption; contact the company's emergency center, since the measures to be taken depends on the proportion of the accident, the characteristics of the body and the quantity of the product involved. Spilled product should not be reused.

Preventing secondary hazards: Prevent product from contaminate streams, lakes, sources of water, sewage and effluents.

7. HANDLING AND STORAGE

- **Handling:**
- **Technical measures:** The product DIABENTIURON GENBRA 500 SC is an acaricide by contact and ingestion, recommended for pest control in cotton, coffee, citrus, beans, watermelon, melon, cabbage, rose, soy and tomato. Before using the product, please read the label, package leaflet and the prescription. THE USE OF INDIVIDUAL PROTECTION EQUIPMENT IS MANDATORY.
- **Prevention of worker exposure:** use PPE as described in Item 8. Do not eat, drink or smoke during handling of product. When opening the packaging do so to avoid spillage. Do not use personal protective equipment when is damaged and / or defective. Do not unclog nozzles, orifices, pipes and valves with the mouth. Do not handle and / or charge damaged packaging.
- **Precautions for safe handling:** use PPE as described in 8. Where possible, keep the product in packages and in indoor settings.
- **Guidelines for safe handling:** use PPE as described in Item 8. Handle the product with appropriate local exhaust ventilation or in a well-ventilated area. In case of poisoning symptoms, immediately stop working and proceed as described in Item 4 of this sheet.
- **Hygiene measures:**

  Appropriate: Take a bath and change clothes immediately after using the product. Wash contaminated clothing separately, avoiding contact with other utensils for personal use. Wash hands before eating or smoking. Do not handle this material near food, feed or drinking water.

  Inappropriate: Do not wash contaminated clothing together with other pieces of clothing or utensils for personal use.
- **Storage**

  Technical measures

  Appropriate: Keep the product and any leftovers in their original containers, properly closed.

  Inappropriate: Avoid direct exposure to sunlight.

  Storage conditions

  Suitable: Keep the product in its original packaging, always closed. The site must be unique to guard toxic products and must not be in contact with food, beverages, feed or other materials. The place should be ventilated, covered and have waterproof floor. Put a warning sign with the words: WARNING POISON. Lock the place, avoiding unauthorized access, especially for children. There should always be adequate packaging available to wrap broken packaging or to collect leaked products. In the case of warehouses, the instructions in NBR 9843 of the Brazilian Association of Technical Standards – ABNT should be followed. Observe the provisions of state and municipal legislation.

  To avoid: Humid places with sources of heat.
- **Incompatible products and materials:** do not store together with food, beverages, including those intended for animals.

- **Safe packaging materials**
  Recommended: product already packed in appropriate packaging.
  Not suitable: do not remove the product from its original packaging.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

- **Engineering control measures:** Provide adequate ventilation. The operator respiratory protection equipment should always be used even in case of good ventilation. Keep containers tightly closed.
- **Specific control parameters:**
  Occupational exposure limits:

Common name	Exposure Limit	Type	Effect	Reference
Diafenthiuron	Not established	TLV-TWA	---	ACGIH 2014
		REL-TWA	---	NIOSH
		PEL-TWA	---	OSHA
Ethylenglycol	Not established	TLV-TWA	---	ACGIH 2014
		REL-TWA	---	NIOSH
		PEL-TWA	---	OSHA

Biological Indicators

Common name	Biological Limit	Type	Notes	Sampling time	Reference
Diafentiuron	Not established	BEI	----	----	ACGIH 2014
Ethylenglycol	Not established	BEI	----	----	ACGIH 2014

- **Equipments for individual safety:**
  Respiratory protection: Use mask with combined filter.
  Hand protection: Wear gloves.

Eye protection: Wear safety glasses with side shields.

Skin and body protection: Wear water-repellent cotton overalls with long sleeves over the glove cuffs and the legs of the trousers over rubber boots, Arab cap and waterproof apron.

Special Precautions: Keep PPE clean and perform properly periodic inspections and possible maintenance and / or replacements of damaged equipment.

9. PHYSICAL AND CHEMICAL PROPERTIES

- **Physical state**: liquid.
- **Form**: concentrated suspension.
- **Appearance**: opaque.
- **Color**: Beige.
- **Odor**: characteristic.
- **pH**: 7.96 in 1% aqueous m / v sol (19.8-20.1 ° C).
- **Melting point**: Not available.
- **Initial boiling point and boiling point range**: not available.
- **Flash Point**: Not available.
- **Flammability**: Not available.
- **Evaporation rate**: not available.
- **Lower / upper flammability or explosive limits**: not available.
- **Vapor pressure**: not available.
- **Vapor density**: not available.
- **Density**: not available.
- **Solubility**: soluble in water.
- **Partition coefficient**: n-octanol / water: not available.
- **Auto-ignition temperature**: not available.
- **Decomposition temperature**: Not available.
- **Viscosity**: 537.2 mPa.s at 20 ° C and 379.2 mPa.s at 40 ° C.
- **Corrosivity**: the product presented corrosion rate for stainless steel = 0.0011 mm / year, aluminum = 0.0192 mm / year, copper = 0.0126 mm / year and 0.0215 iron mm / year. The brass plates showed no signs of corrosion when in contact with the test substance.
- **Surface tension**: not available.

10. STABILITY AND REACTIVITY

- **Chemical stability:** product is stable under indicated conditions of use and storage.
- **Reactivity:** no data available on product reactivity.
- **Possibility of hazardous reactions:** no data available.
- **Conditions to avoid:** Avoid contact with heat, high temperatures, sources of ignition and exposure to sunlight.
- **Incompatible materials or substances:** No data available.
- **Hazardous decomposition products:** Prolonged or repeated exposure may cause toxic gases and irritants.

11. TOXICOLOGICAL INFORMATION

- **Acute toxicity:**

  Oral LD50 (rats): 500 mg / kg.
  LD50 Dermal (male and female rats) :> 2000 mg / kg
  LC50 Inhalation (male and female rats) (4h): 2.05 mg / L

- **Local Effects:**

  Skin irritation: Diafenthiuron: not irritating to the skin of rabbits. Ethylene glycol: Irritating to skin.
  Eye irritation: Diafenthiuron: Moderately irritating to rabbit eyes. Ethylene glycol: irritating to eyes.
  Skin Sensitization: No data available.
  Respiratory sensitization: No data available.

- **Chronic toxicity:**

  Germ cell mutagenicity: Diafenthiuron: No data available. Ethylene glycol: negative results in genotoxicity test in in vitro an in vivo.
  Carcinogenicity: Diafenthiuron: The substance is known to be non-carcinogenic. Ethylene glycol: studies in rats and mice did not reveal any potential carcinogenic.
  Reproductive toxicity: No data available.
  Systemic toxicity to target organ - single exposure: no data available.
  Systemic toxicity to target organ - Repeated exposure: no data available available.
  Aspiration hazard: no data available.
  Main Symptoms: Ingestion of large quantities of the product may cause such as nausea, vomiting, colic and diarrhea. Direct contact with eyes or skin may cause redness and discomfort at the site of contact.

12. ECOLOGICAL INFORMATION

- **Environmental, behavioral and product impacts:**

- **Persistence / degradability:** no data available.

- **Eco toxicity:**

  Diafenthiuron:
  Toxicity to fish: Rainbow trout: LC50 (96h): 0.0007 mg / L.
  Microcrystalline toxicity: Daphnia**:** EC50 (48h): 0.5 mg / L.

  Ethylene glycol:
  LC50 for fish (96h): 54700 mg / L.
  EC50 for micro crustaceans (48h): 41000 mg / L.

- **Mobility in soil**: No data available.
- **Bioaccumulation**: no data available.

13. CONSIDERATIONS ON FINAL DESTINATION

- **Methods of treatment and disposal**:

Product: if this product becomes unfit for use consult the registrant through the telephone number indicated on the label for return and final destination of the product. The expired products should be appropriately discarded.

Remains of products: keep any leftovers from the product in their original packaging and properly closed.

Used packaging: Within one year from the date of purchase, it is mandatory the return the empty packaging to the establishment from where it was purchased. If the product has not been fully used within that period, and still is within its validity period, it will be possible to return the packaging within 6 months after the expiration date. The user must be kept the returning receipt for inspection purposes for at least one year.

14. TRANSPORT INFORMATION

- **National and international regulations:**

TERRESTRIAL TRANSPORTATION: Resolution ANTT 5232 of 12/16/2016 of the Ministry of Transport.

UN Number: 3082
Proper shipping name: SUBSTANCE THAT PRESENTS RISK FOR THE ENVIRONMENT, LIQUID, N.E. (Diafenthiuron)
Risk class: 9
Number of risk: 90
Packing group: III

MARITIME TRANSPORT: IMDG (International Maritime Dangerous Goods Code).
UN number: 3082
Name and description: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, WE. (Diafenthiuron)
Class risk: 9
Packing group: III

15. REGULATORY INFORMATION

- **Regulatory information:**

ABNT NBR - 14725
Resolution 5232 - ANTT
IMDG CODE

16. OTHER INFORMATION

"This MSDS was prepared by TOXICLIN® Medical Services, based on data provided by the Distributor. The information in this MSDS represents the current and accurately reflect our best knowledge for the proper handling of this product in accordance with the specifications on the label and package leaflet. Any other uses other than those recommended, will be under the user's responsibility.

Acronyms:
ABNT - Brazilian Association of Technical Standards
ACGIH - American Conference of Governmental Industrial Hygienists
BCF - Bioaccumulation factor
CAS - Chemical Abstracts Service
EC50 - Effective concentration 50%
LC50 - Lethal concentration 50%
LD50 - Lethal dose 50%
Koc - Partition coefficient normalized by organic carbon
MT - Ministry of Transport
NBR - Brazilian Standard
NIOSH - National Institute for Occupational Safety and Health
OSHA - Occupational Safety & Health Administration
PEL - Permissible Exposure Limit
REL - Recommended Exposure Limit
TLV - Threshold Limit Value
TWA - Time Weighted Average

Subtitles:
Rating impossible - there is not enough data available for classification of the product.

Unclassified - product does not fall within the GHS classification category and, therefore, presents no danger.

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